

ECEN 665

Section – 601

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VCO

Lab 05: (Post-lab) report

Date: 19-03-2026

Texas A&M University

Objectives

- To learn the use of periodic steady state (pss) simulation tools in spectre (cadence) in the characterization of a VCO including its phase noise.
 - To understand the operation of a CMOS LC VCO and analyze its performance trade-offs.
-

1. Schematic Setup

Using the CMOS 0.5um technology library in Cadence, constructed the LC-based CMOS VCO schematic as shown in Figure 1. The circuit is designed to operate around **1.9 GHz**.

Component Values

Transistor Parameters (Table 1)

Transistor	W [um]	L [um]	Multiplicity
N0, N1	9	0.2	8
P0, P1	15	1.2	12
P3, P4	30	1.8	16

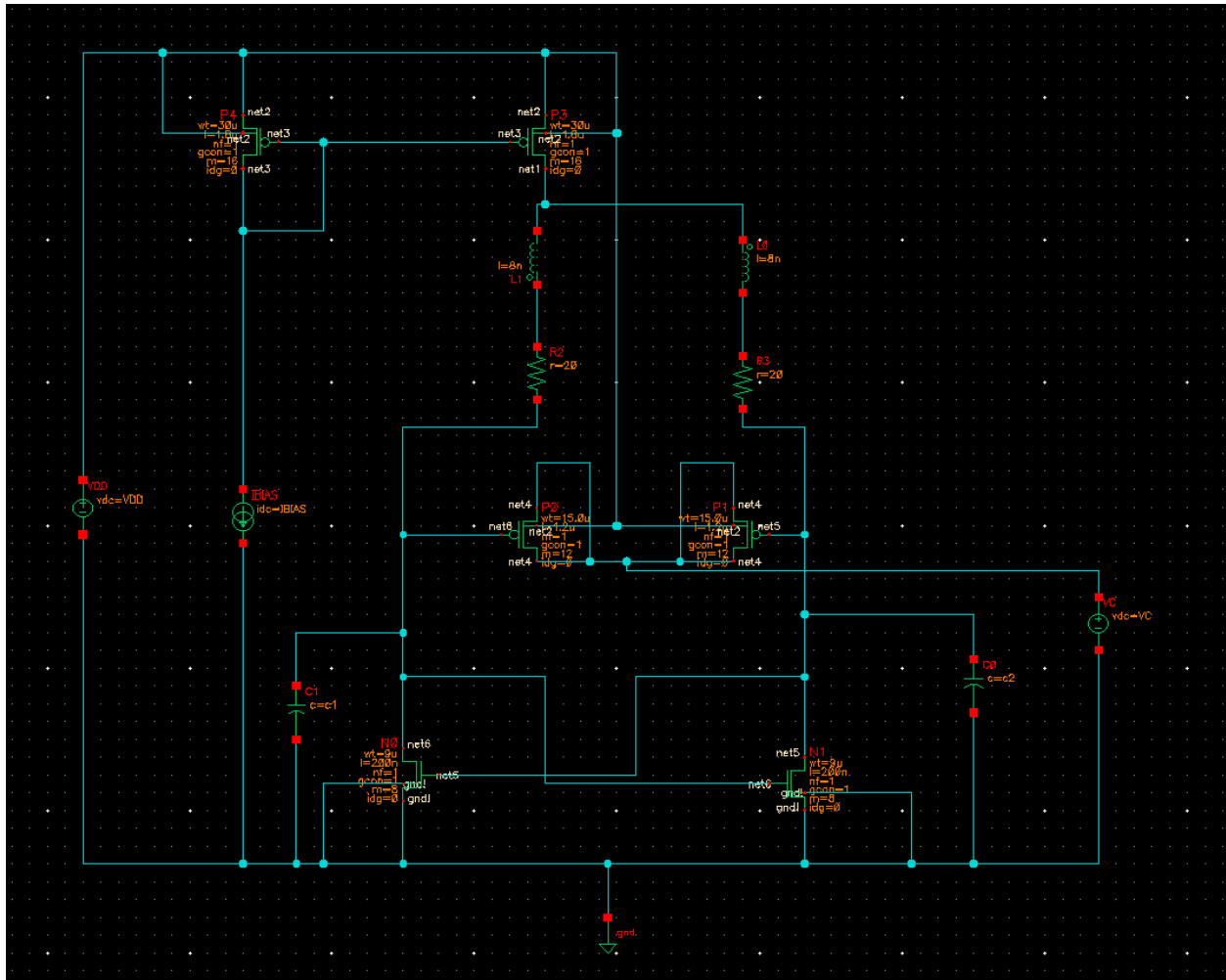
Component Values (Table 2)

Component	Value
L0, L1	8n
R2, R3	20Ω
C0, C1	200fF
IBIAS	1mA
VC	700mV
VDD	3V

Notes:

- LC tank formed by L1, L0 ($Q \approx 5$); resonance determined by R2, R3.
- Varactors are PMOS transistors P0, P1.

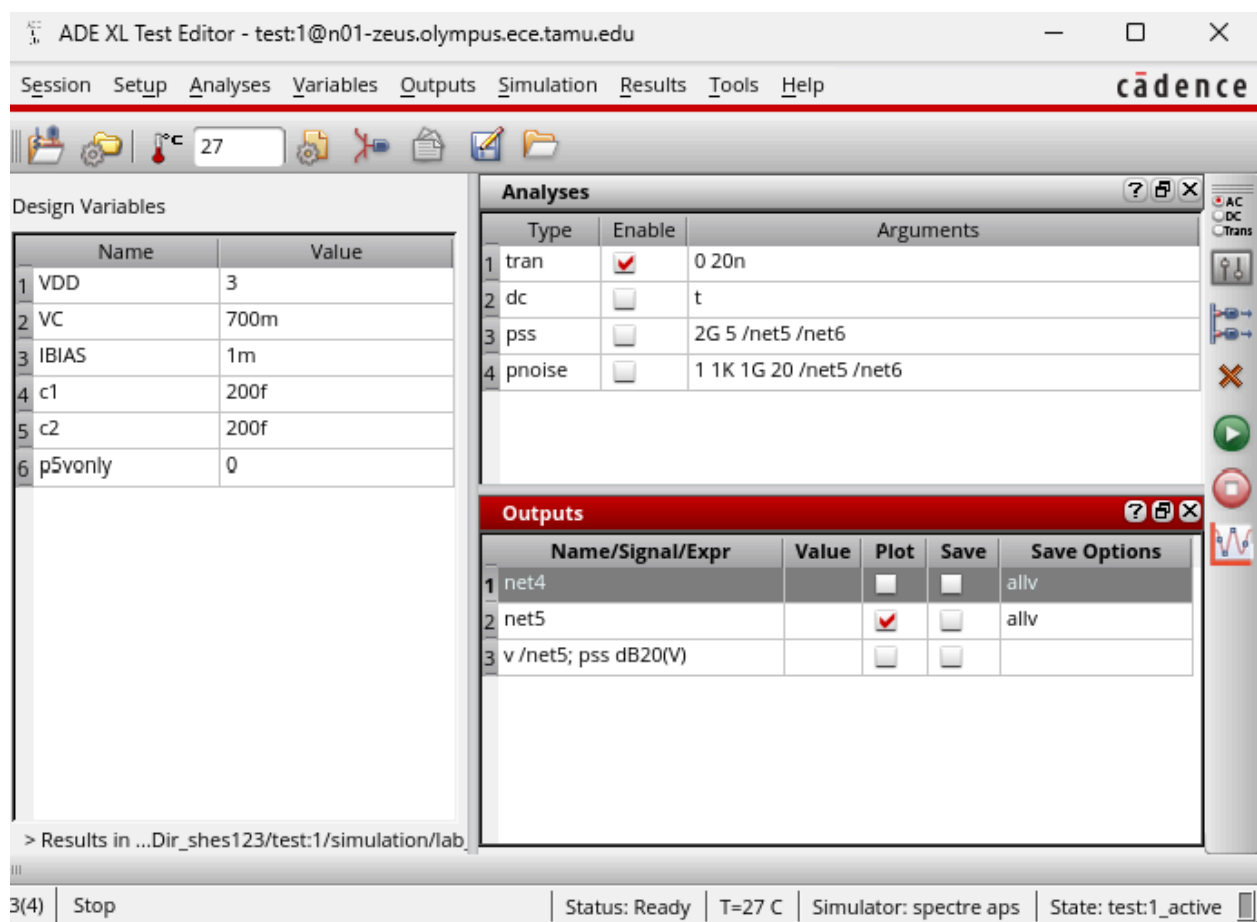
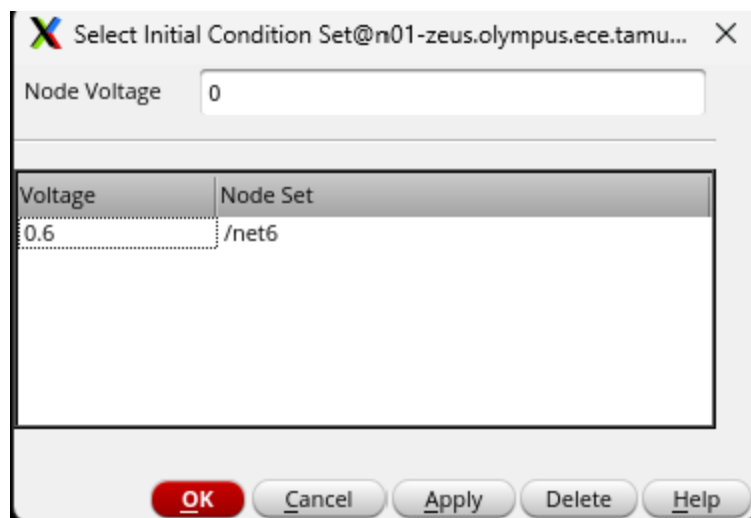
- C0, C1 are load parasitics.
- Negative conductance (oscillation starter) is cross-coupled NMOS N0, N1.

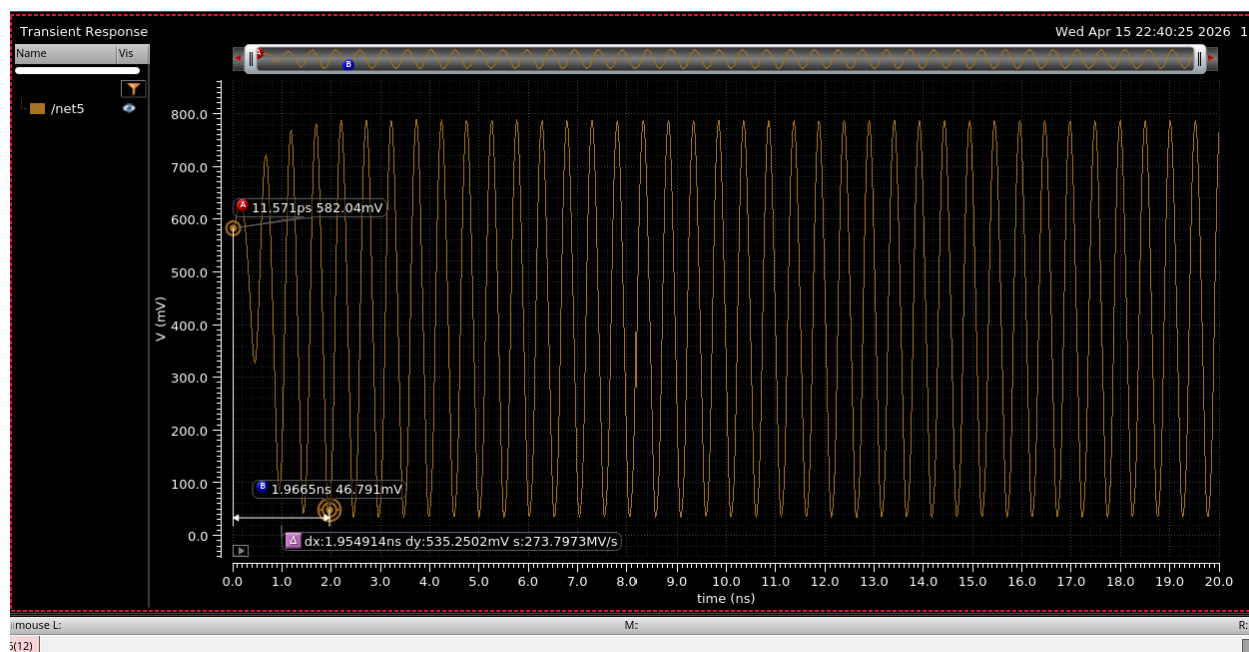
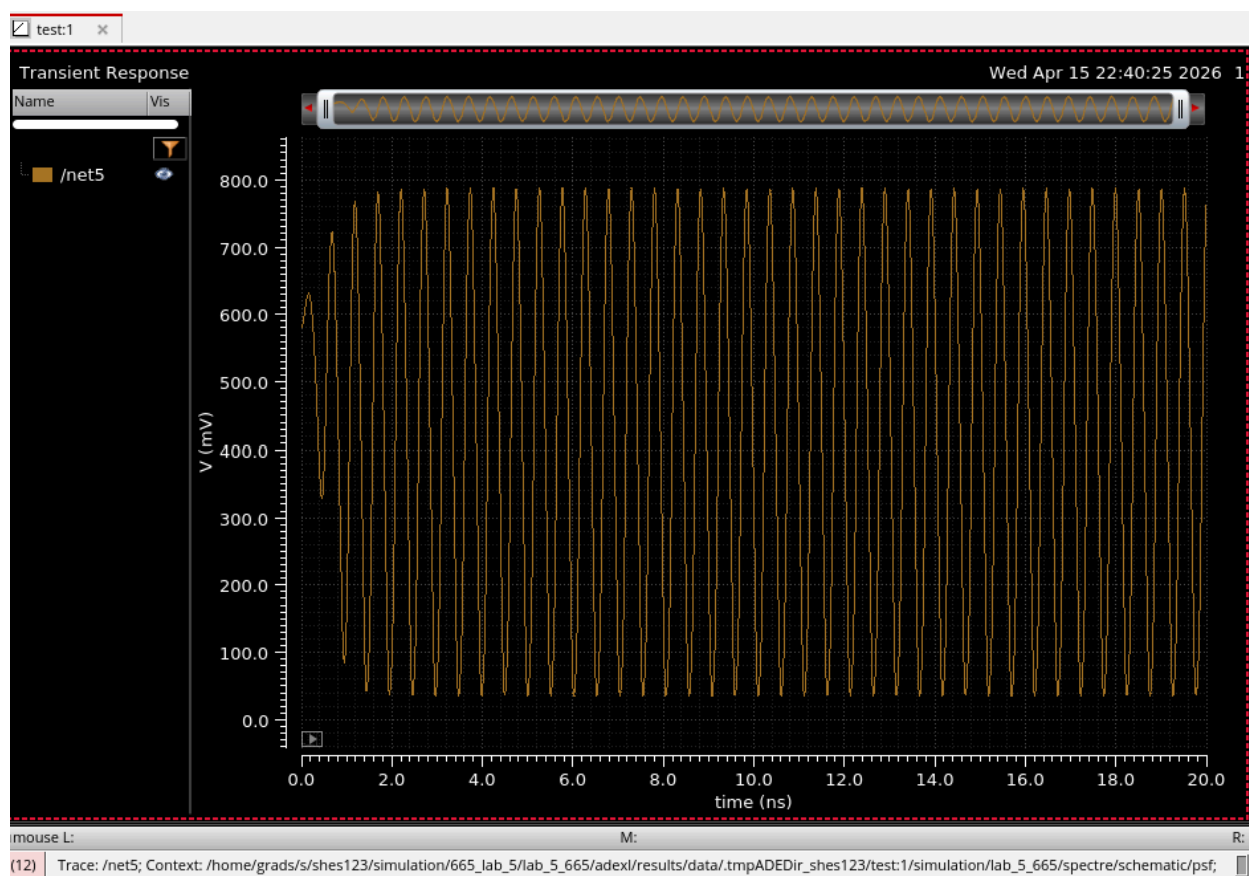


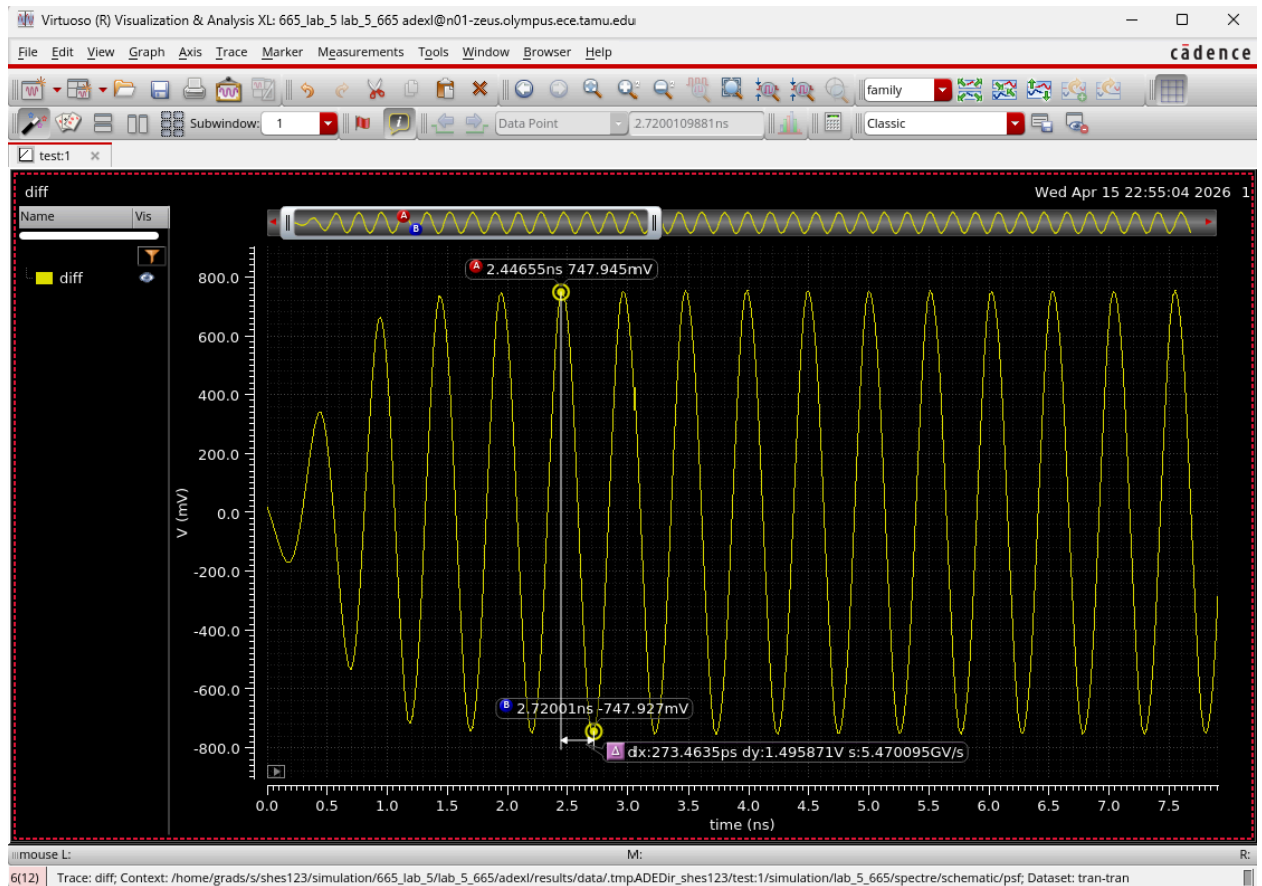
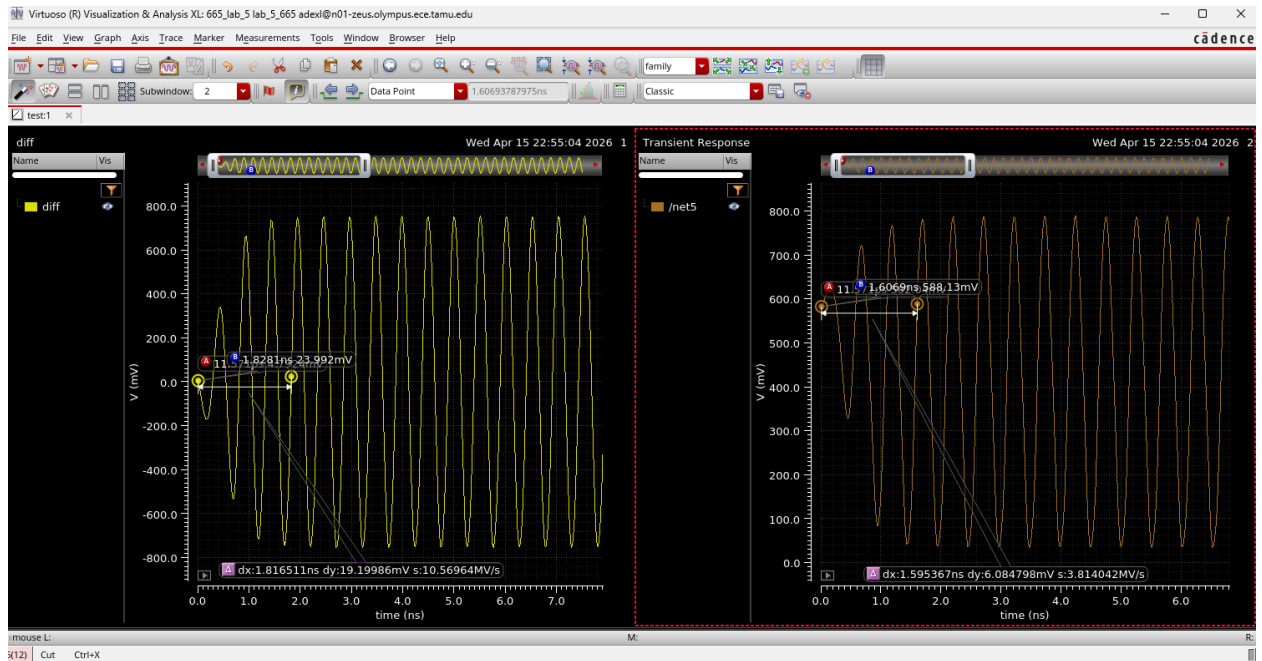
2. Transient Simulation

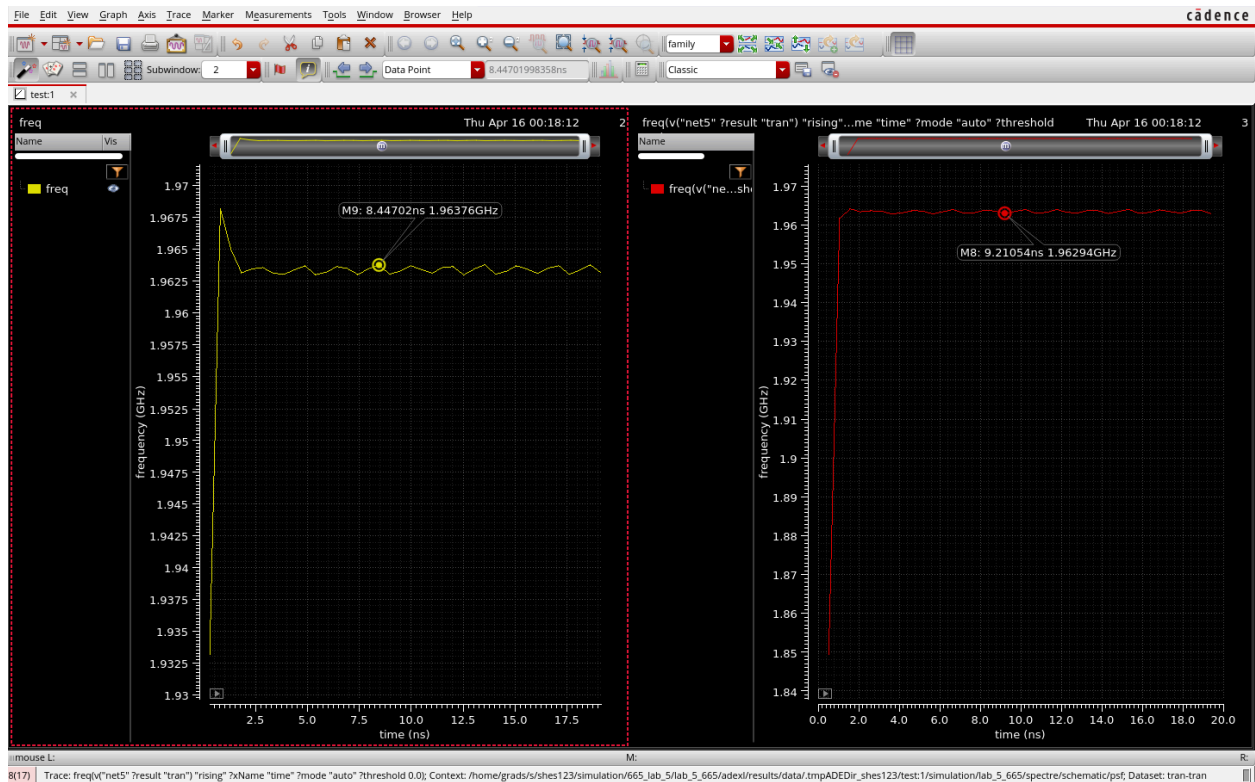
Performed transient analysis over **20ns** to verify oscillation and estimate settling time and output frequency.

- If oscillations do not start: set initial condition via Simulation -> Convergence Aids -> Initial Condition... and set output node offset by few mV.
- **Output:**
 - Output waveform plotted.
 - Frequency measured with calculator.









- Settling time: 1.81 ns
- Output frequency: 1.96 GHz

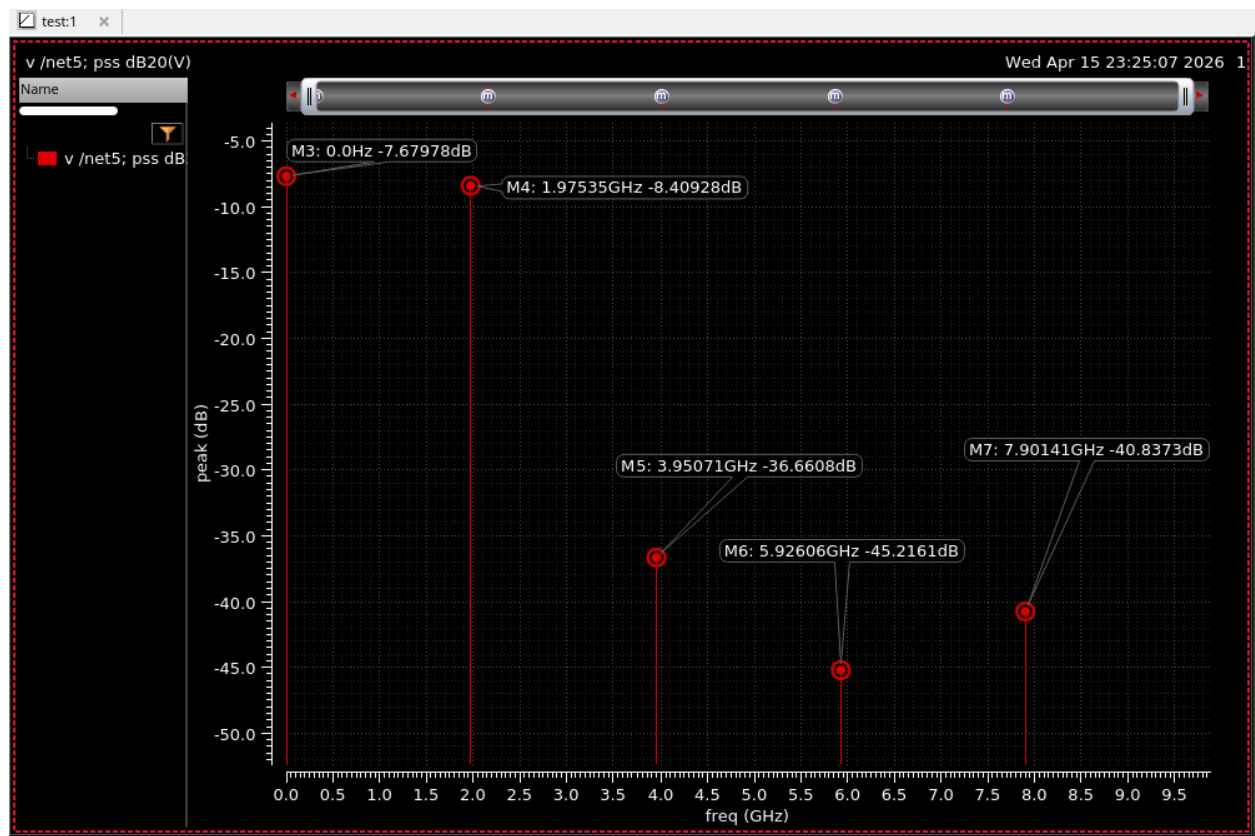
3. PSS Simulation

PSS analysis provides oscillation frequency, spectrum, amplitude, and enables phase noise analysis (pnoise).

Simulation Setup

- Use conservative analysis mode (critical for phase noise accuracy).
- Set stabilization time as measured in transient sim.
- Output nodes: drain/gate of N0, N1.
- Fundamental frequency: starting guess (PSS will correct).

PSS simulation:



Noise output:

Choosing Analyses -- ADE L (1)@n01-zeus.olympus.ece.tamu.edu

☐ qpac ☐ qpnoise ☐ qpxf ☐ qpasp
☐ hb ☐ hbac ☐ hbstb ☐ hbnoise
☐ hbsp ☐ hbxf

Periodic Noise Analysis

PSS Beat Frequency (Hz)

Multiple pnoise ☐

Sweeptype Relative Harmonic

Output Frequency Sweep Range (Hz)

Start Stop

Sweep Type

☒ Points Per Decade
☐ Number of Steps

Add Specific Points ☐

Add Points By File ☐

Sidebands

Method ☒ default ☐ fullspectrum

Maximum sideband

When using shooting engine, default value is 7.

Output

Positive Output Node
Negative Output Node

Noise Type

Timeaverage: single-sided spectrum and harmonic-referred (modulated) noise analysis

Contribution Type:

☐ USB ☐ AM ☐ PM ☐ AM&PM ☒ ALL(AM,PM,USB,LSB)

For fmjitter, PM noise must be enabled

Noise Separation ☐

Separate noise into source and gain

Generate Osc Macro Source ☐

Enabled ☒

Direct Plot Form@n01-zeus.olympus.ece.tamu....

Plotting Mode: Append

Analysis

☐ pss ☒ pnoise

Noise Type

☐ USB ☐ LSB ☐ AM ☒ PM

Function

☐ Output Noise ☐ -20dB/dec Line

☐ Jc ☐ Jcc

☒ Phase Noise

Add To Outputs ☐ Plot

> Press plot button on this form...

Close Help

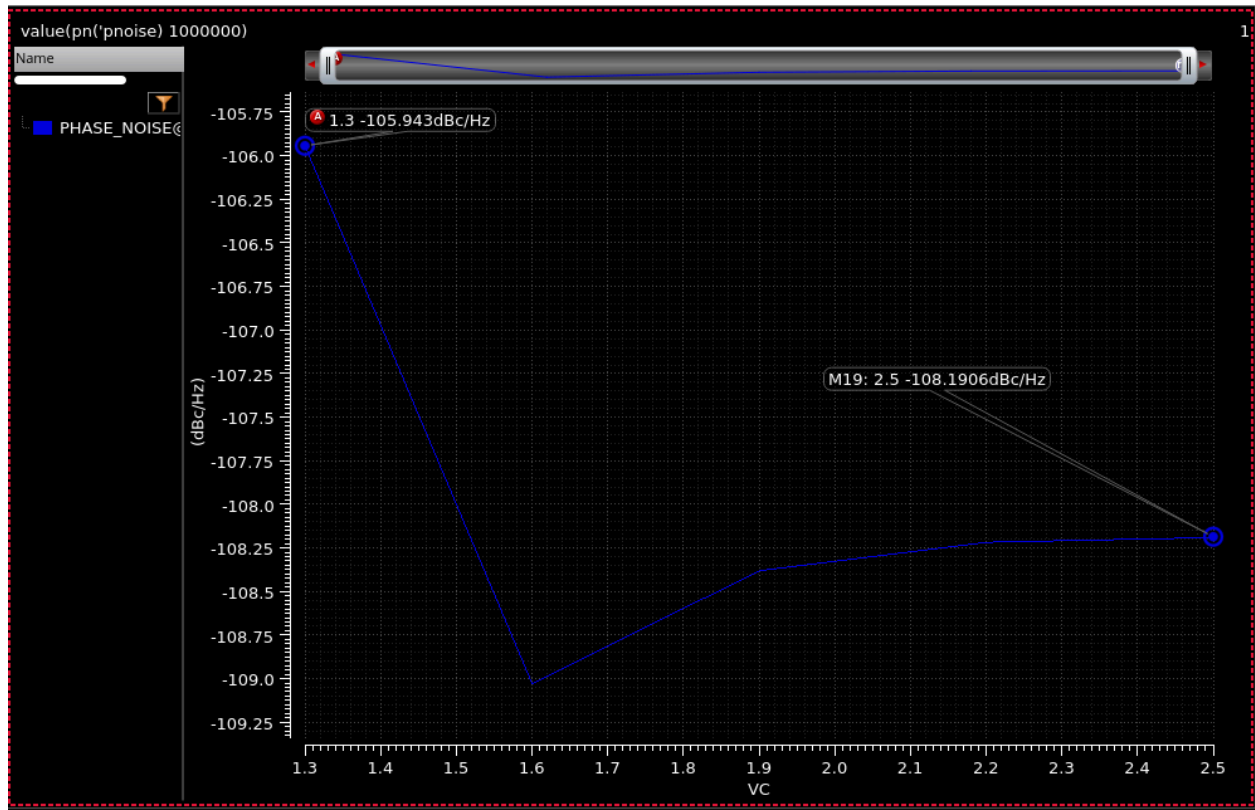
Tue Apr 14

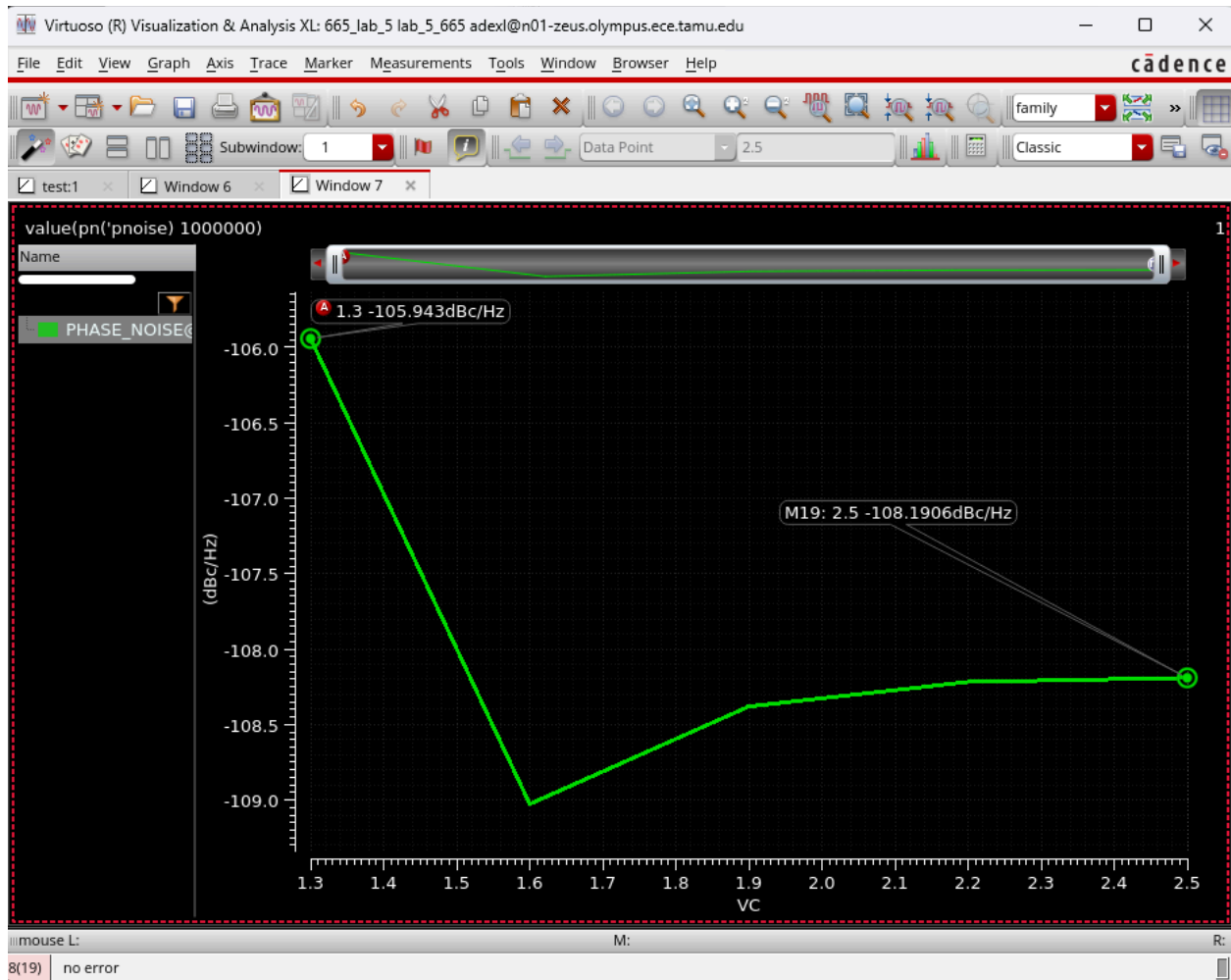
4. Parametric Simulation

To study VCO behavior with varying tuning voltage (VC):

- Set VC as a sweep variable in range ____ 1.3 TO 2.5 ____.
- Run both transient and PSS simulations.
- Use calculator "family" button for parametric curves:
 - 'frequency', 'ymax', 'value at 1MHz' (for phase noise).

(Insert Figures: Frequency vs VC, Output amplitude vs VC, Phase noise vs VC)





5. VCO Analysis

5.1 Role of Varactors

- Varactors provide **tuning range** (control oscillation frequency via VC).
- Affect **quality factor (Q)** of tank (parasitic losses can reduce Q).
- Govern **linearity of gain** (capacitance-voltage characteristic).
- *Well-designed varactors maximize tuning range, maintain high Q, and linearity.*

5.2 Design Variable Effects

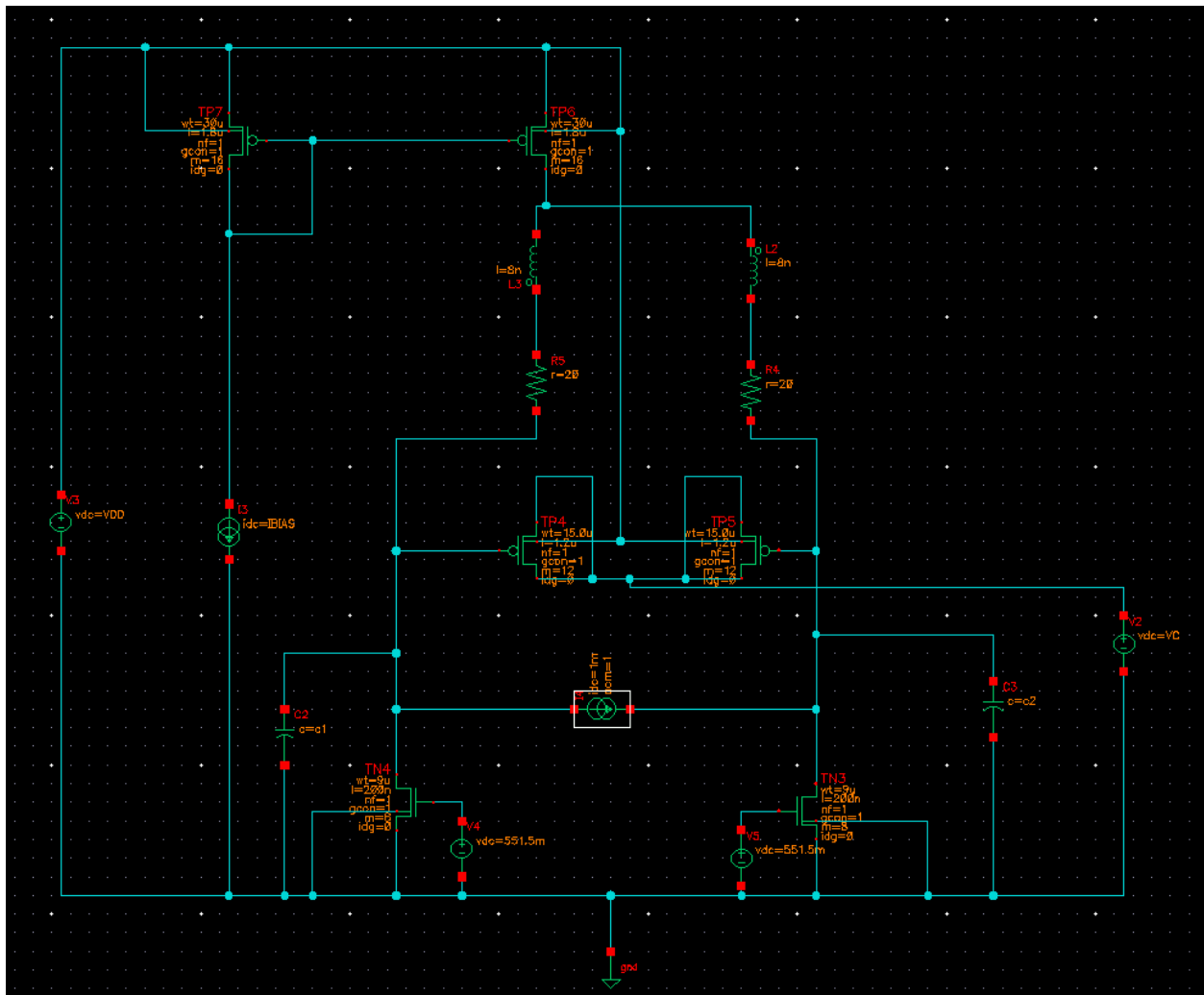
- **Bias Current:**

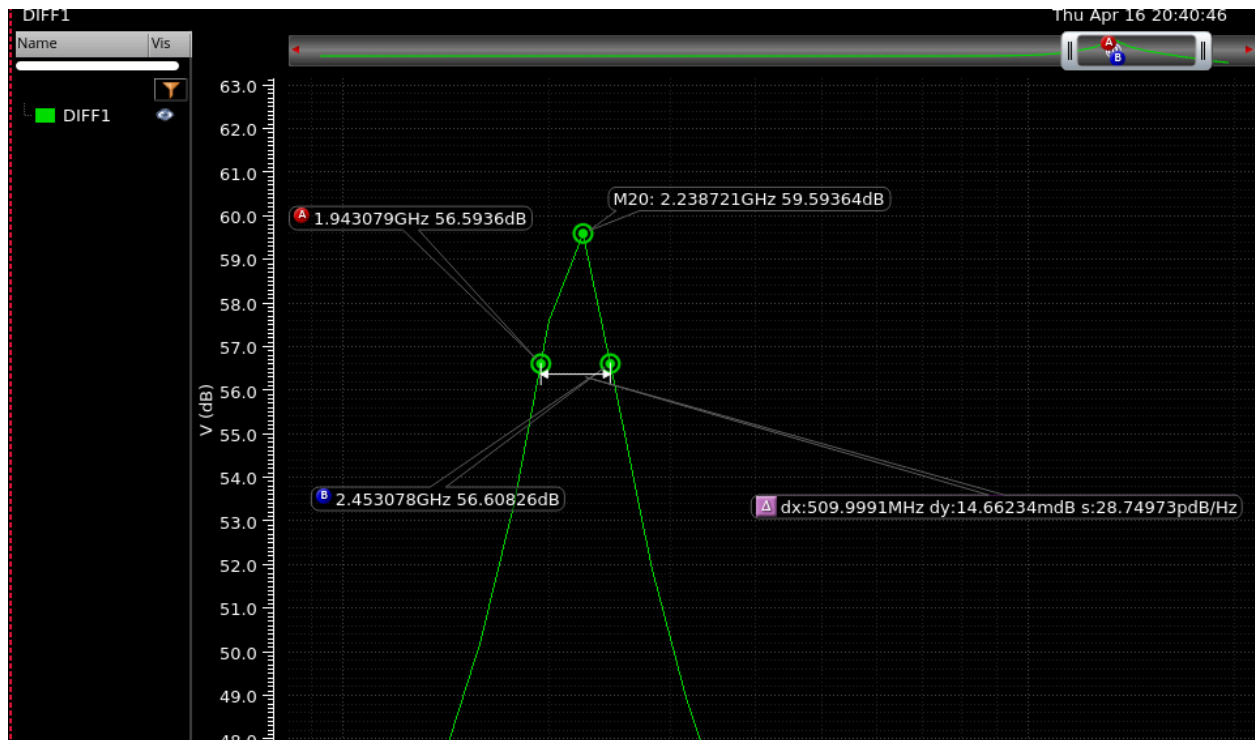
- Higher bias increases oscillation amplitude; improves signal swing; also impacts phase noise negatively if excessive (thermal noise).
- **gm (cross-coupled NMOS):**
 - Larger gm improves negative resistance, enhances oscillation startup/amplitude; excess can lead to higher phase noise.
- **Q factor (LC tank):**
 - Higher Q minimizes losses, improves amplitude and reduces phase noise.
- **NMOS W/L tradeoff:**
 - Increasing W/L increases gm, but also gate capacitance (slower circuit, higher power). Tradeoff between amplitude, phase noise and speed.

5.3 Change Q Factor (from 5 to 9)

THE Q FACTOR MEASUREMENT:

Circuit Configuration to get Q-factor:





Q-factor calculation=peak frequency/(bandwidth)
=(2.23)/(0.5)=4.46 ≈5

Phase noise:

ADE XL Test Editor - test:1@n01-zeus.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

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Design Variables

Name	Value
1 VDD	3
2 VC	700m
3 IBIAS	1m
4 c1	200f
5 c2	200f
6 p5vonly	0

Analyses

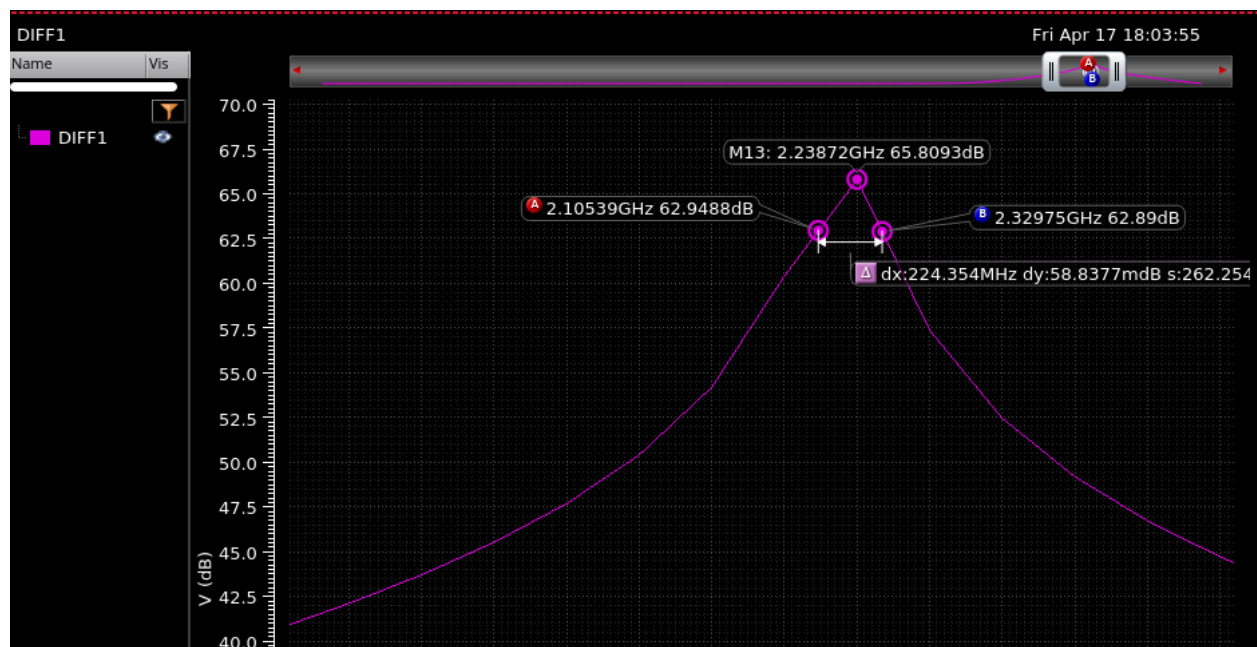
Type	Enable	Arguments
1 tran	☐	0 20n
2 dc	☐	t
3 pss	☒	2G 5 /net5 /net6
4 pnoise	☒	1 1K 1G 20 /net5 /net6
5 ac	☒	1 50G 20 Logarithmic Points Per Decade Start-Stop

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
8 Phase Noise; dBc/Hz, Relati...	wave	☒	☐	
9 (CorFreq:PN-3dB)	(586....	☒	☐	
10 FREQ		☒	☐	
11 YMAX		☒	☐	
12 2 TIMES		☒	☐	
13 PHASE_NOISE@1	-114	☒	☐	
14 DIFF1		☐	☐	

> Results in ...Dir_shes123/test:1/simulation/lab...

7(13) | Load State ... | Status: Ready | T=27 C | Simulator: spectre aps | State: test:1_active

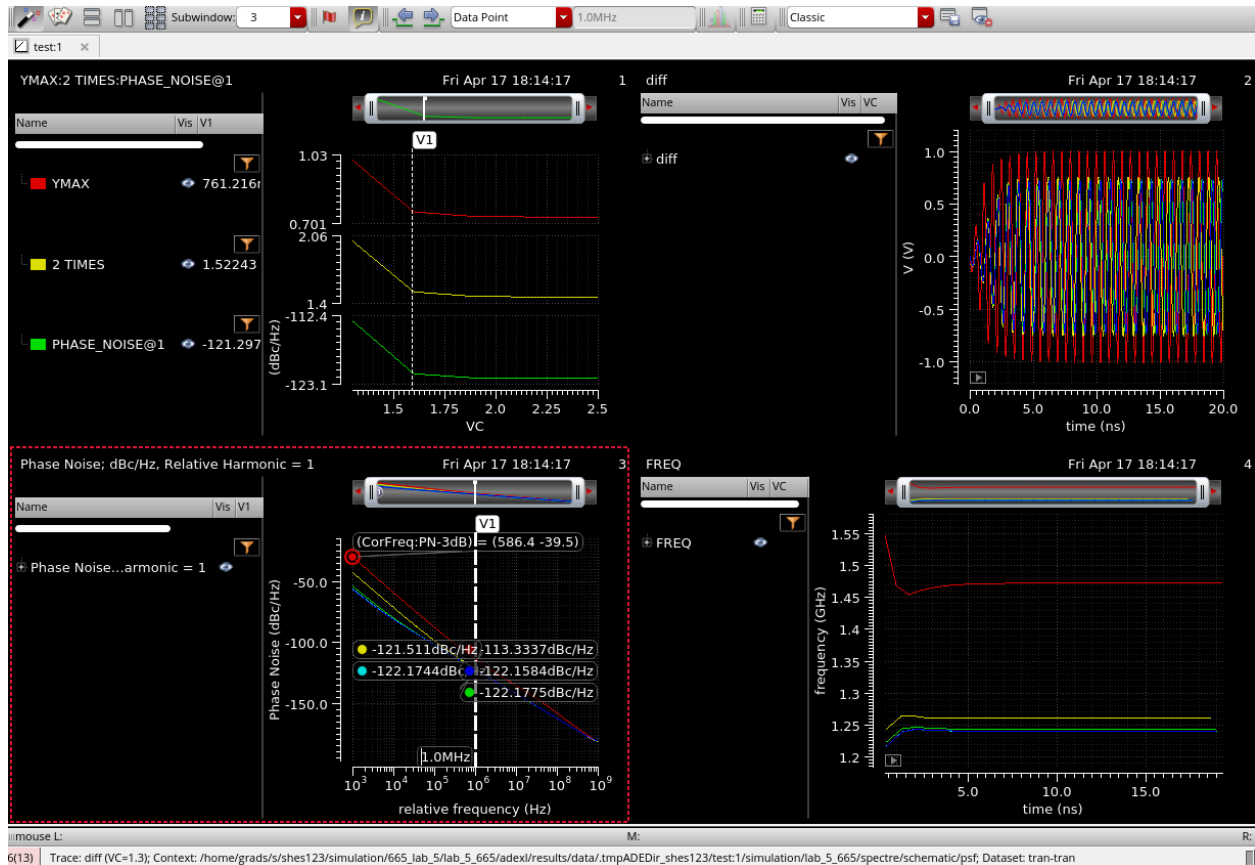


Q factor comes out to be 9.9

Changed the Resistor value to 6 to improve the Q factor

- Simulate with inductor Q=9, obtain parametric plots.
 - Compare phase noise reduction: Δ Phase noise = -113.8457 dBc @1MHz
 - Adjust bias current to match phase noise as Q=5: Required IBIAS = _____
 - Typically, higher Q permits lower bias current for same phase noise.

Running all od the sweep again:



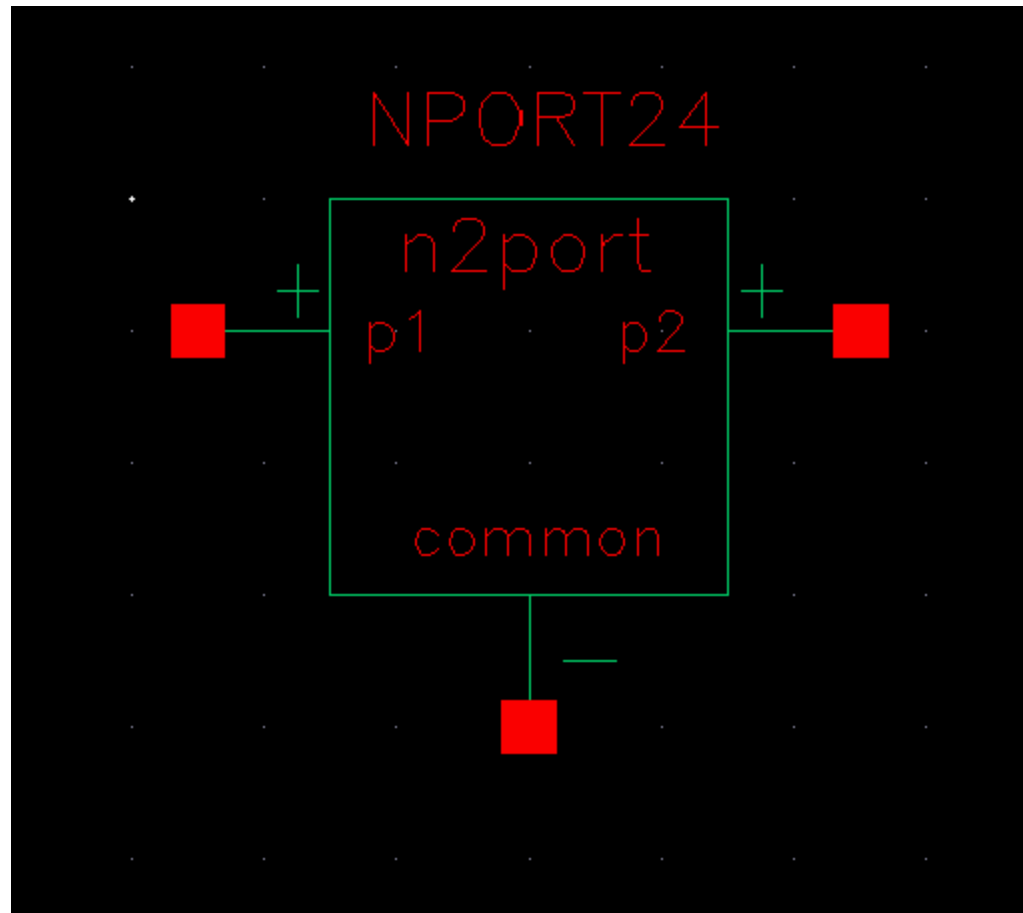
5.4 Measure Q of Tank

- Q measured from simulation: Q = 9.9
- gm, bias current from DC sim: gm = _____, IBIAS = 1mV _____
- Estimate phase noise @1MHz offset (Leeson's formula):
 - PN = -113 dBc/Hz (@1MHz)
- Compare with simulation results: Simulated PN = _____ dBc/Hz

5.5 ASITIC Inductor Design

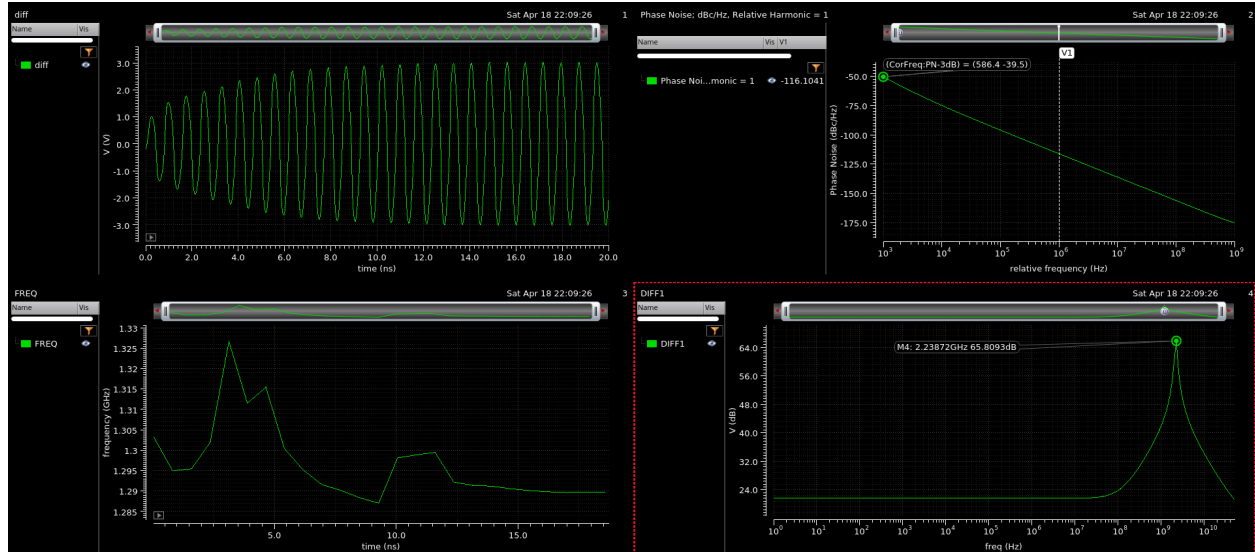
- ASITIC used to design inductor for 1.9 GHz application.
 - Extracted model Q = _____

Importing asitic model from sonnet:



Noise values:

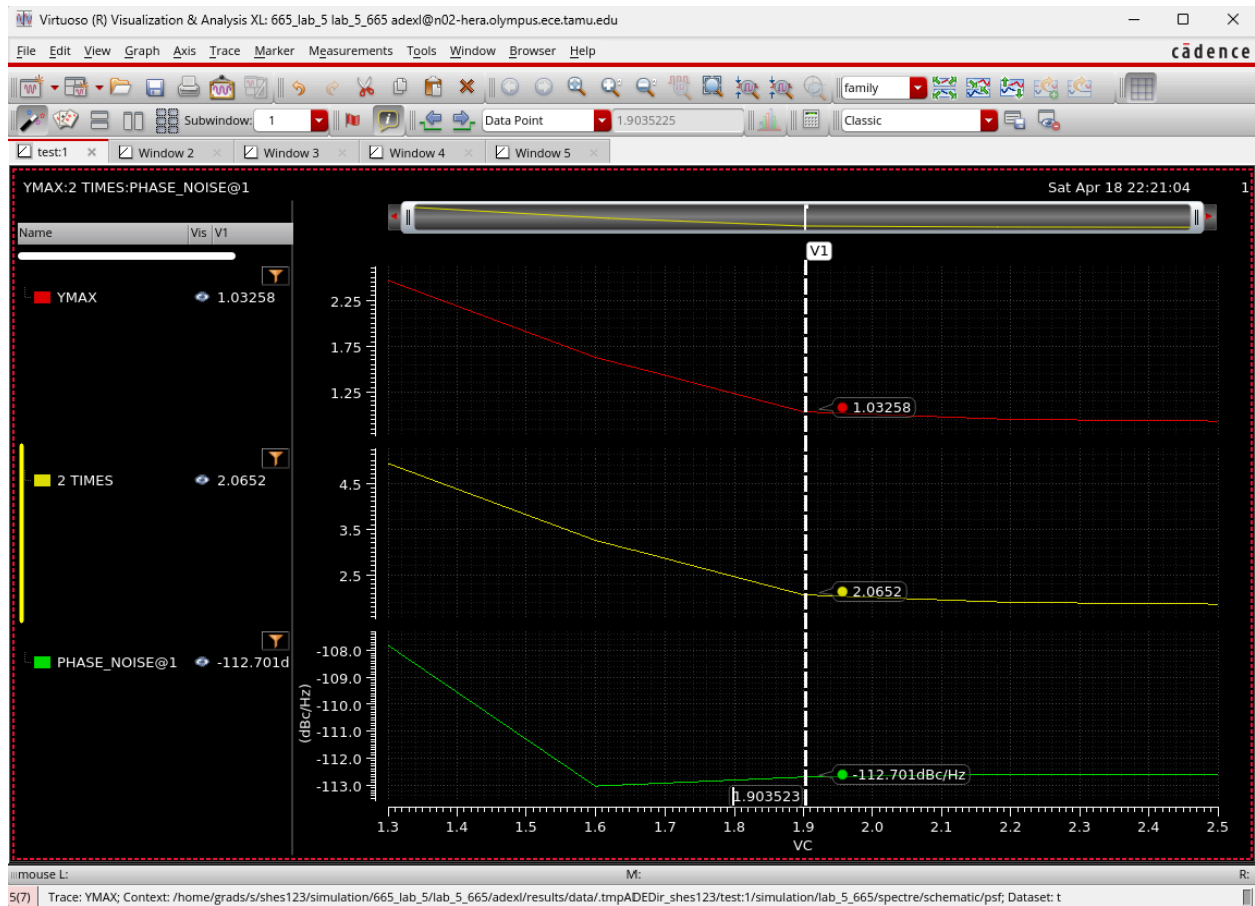
Outputs				
	Name/Signal/Expr	Value	Plot	Save
6	net6		☐	☐
7	et5		☐	☐
8	Phase Noise; dBc/Hz, Rel...	wave	☒	☐
9	(CorFreq:PN-3dB)	(586...	☒	☐
10	FREQ	wave	☒	☐
11	YMAX	3.006	☒	☐
12	2 TIMES	6.013	☒	☐
13	PHASE_NOISE@1	-116.1	☒	☐
14	DIFF1	wave	☒	☐



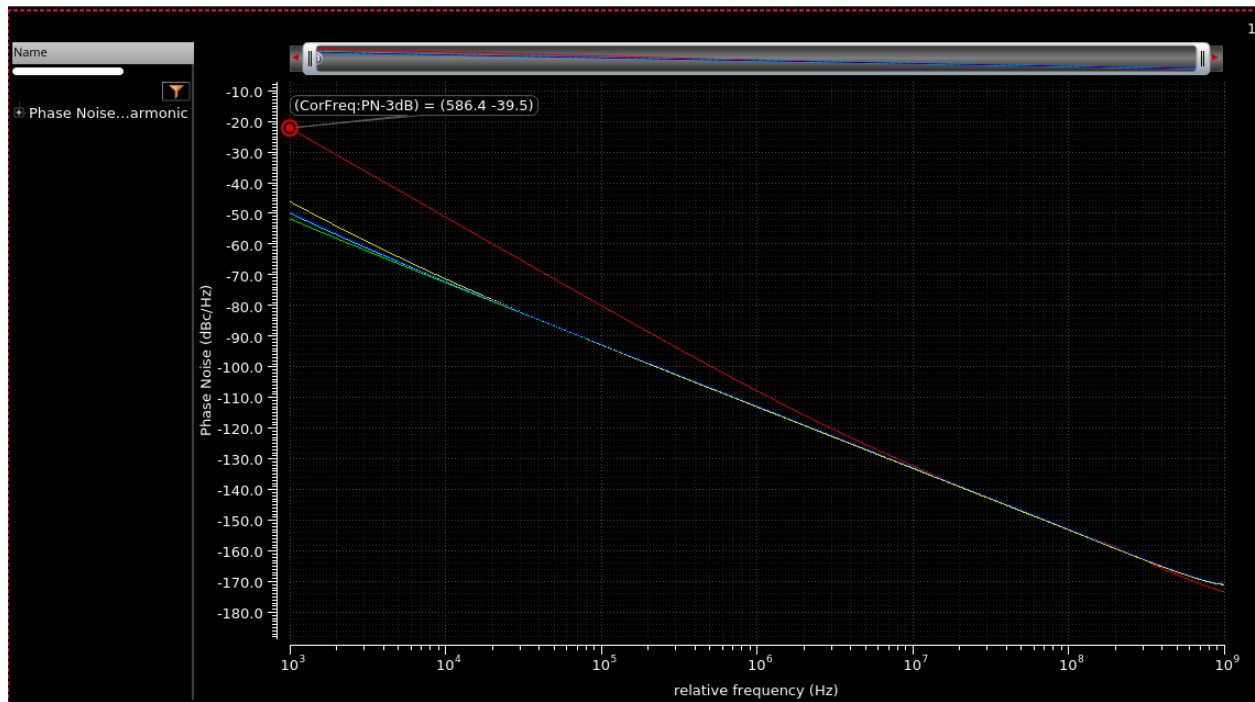
- Re-designed VCO to achieve phase noise of -119 dBc @1MHz offset, 1.9GHz output.
 - Characterization (frequency, amplitude, phase noise, tuning range).

Parametric Sweep of the Voltage VC:

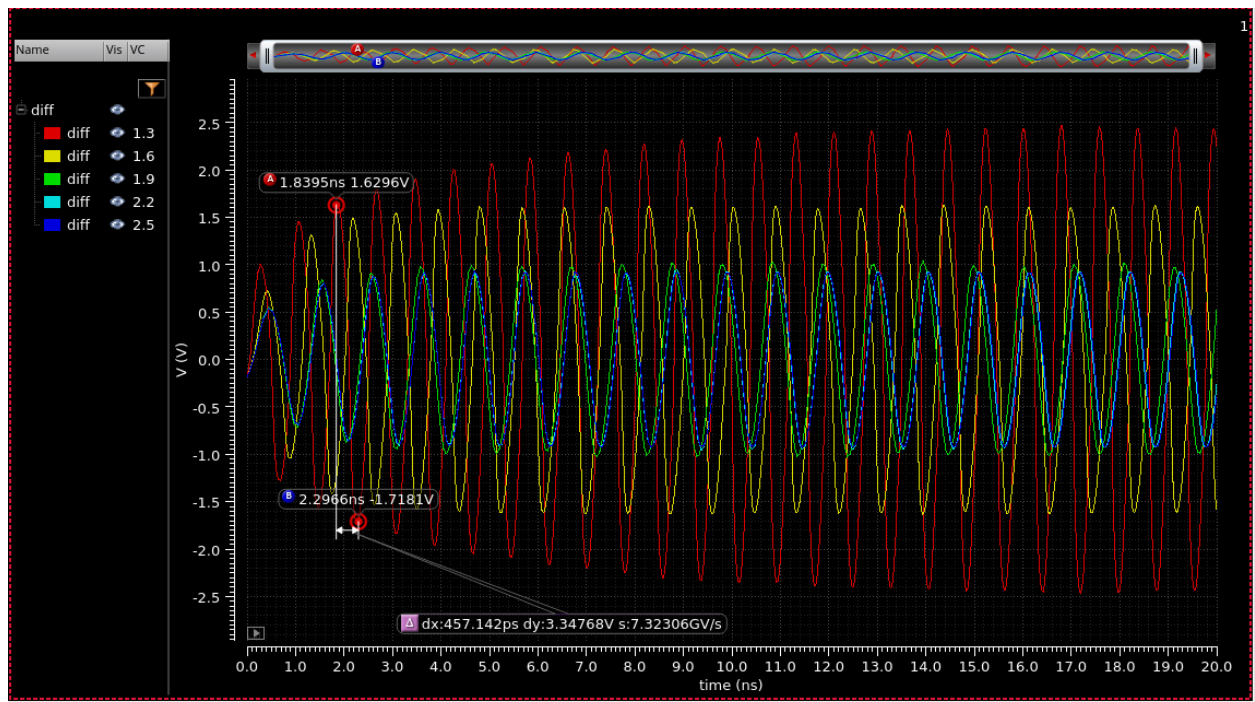
Ymax and phase noise:

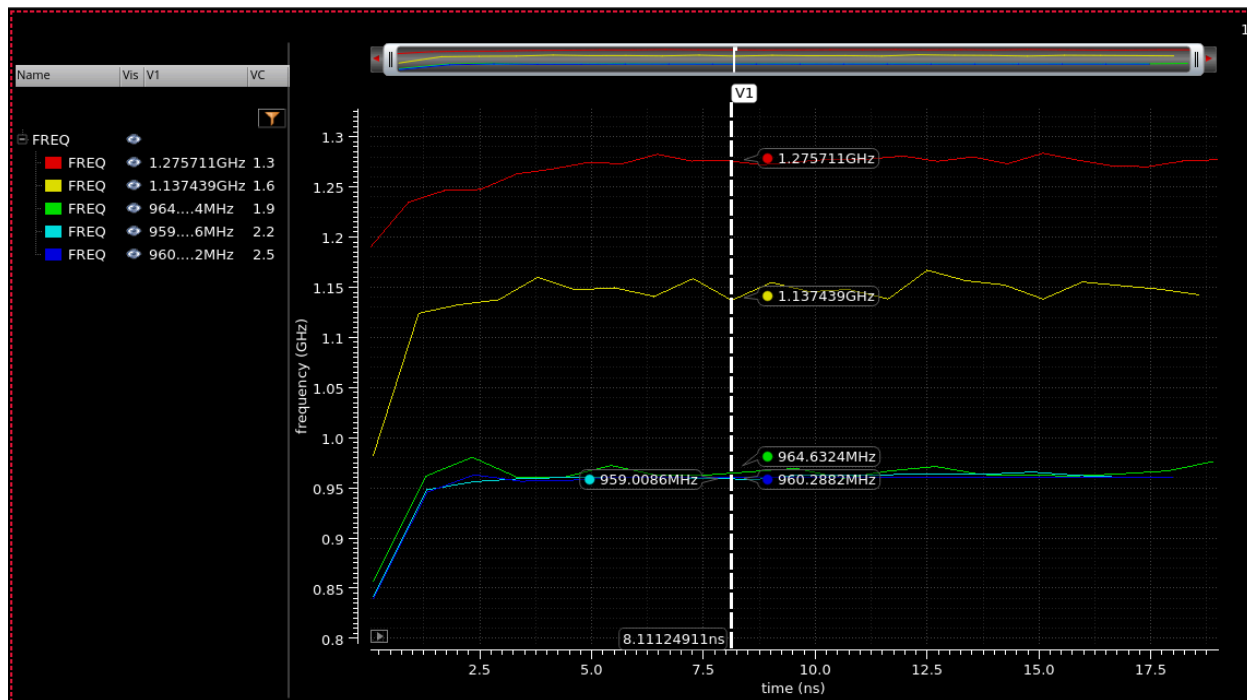


Phase noise vs frequency:

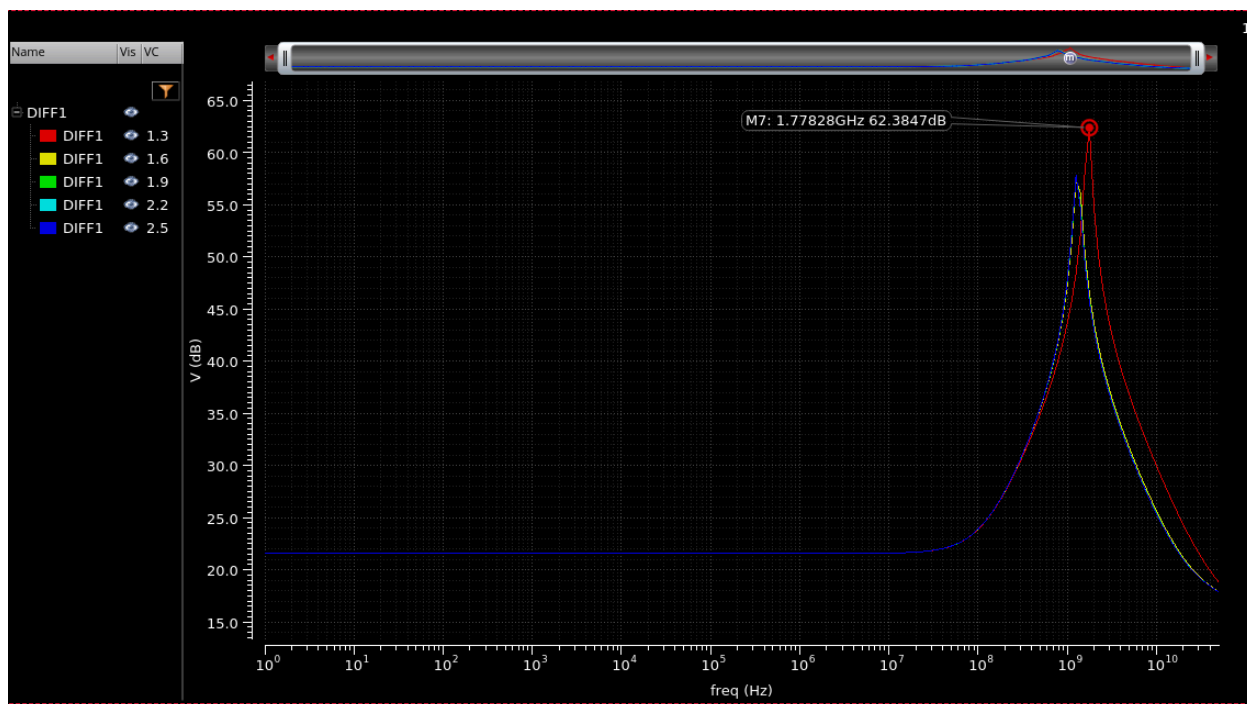


Output swing:(vop-von):

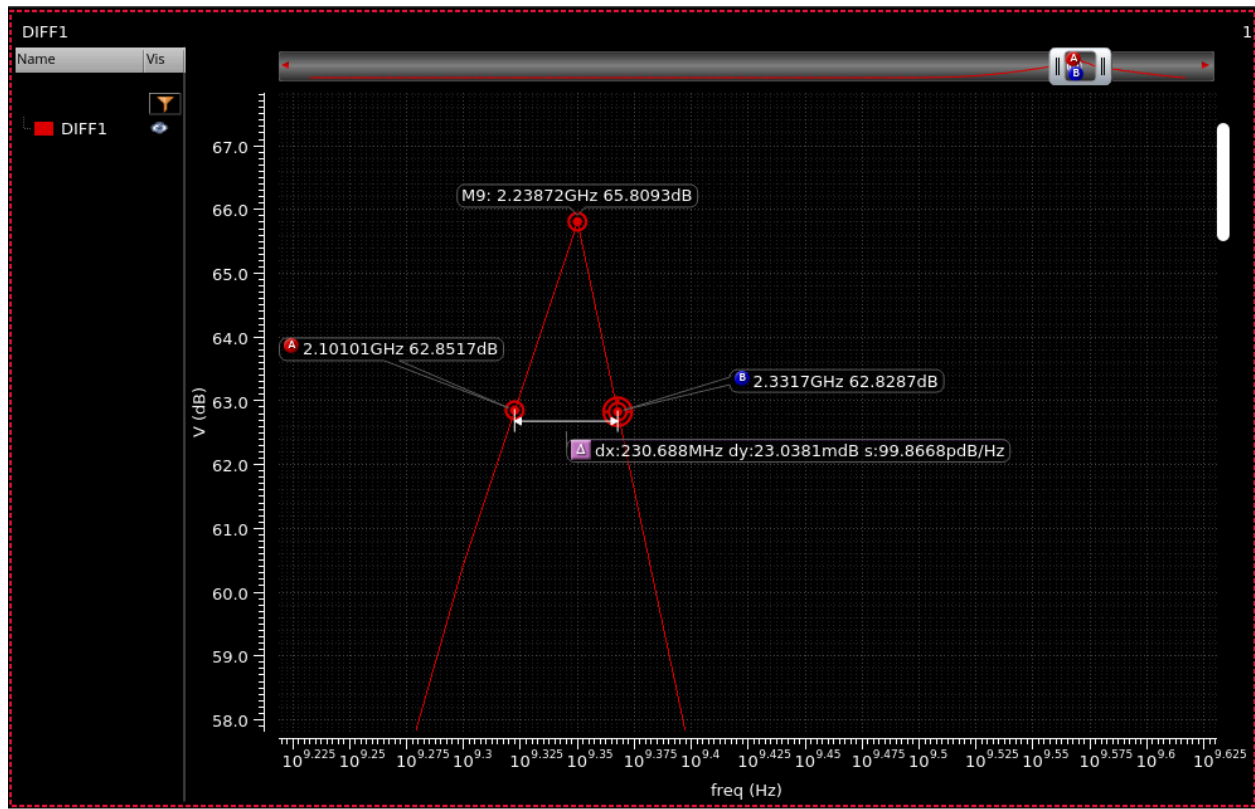




Q factor AC Sweep:



Q factor:



Q factor=(Peak Frequency/(difference between the -3db frequency))

Q=9.7
